

Coping with Emotion Coping: A Corpus to Model Emotions in Text Based on Role Playing

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Abstract

Observing a mismatch between psychological and computational studies on emotions (the ones documenting the internal mechanisms of these phenomena, and the others simplifying them into labels), recent research is bridging the two fields. Many emotion fundamentals, however, are still under-explored in natural language processing. To help reduce this gap, we follow theories on coping, and treat emotions as strategies to cope with salient situations (i.e., how people deal with emotion-eliciting events). This approach allows us to investigate the link between emotions and behavior, which also emerges in language. We introduce the task of coping identification, together with a corpus to do so, constructed via role-playing. We find that coping strategies realize in text even though they are challenging to recognize, both for humans and automatic systems trained and prompted on the same task. We thus open up a promising research direction to enhance the capability of models to better capture emotion mechanisms from text.

1 Introduction

In the last decades, natural language processing (NLP) has tackled the interaction between language and emotions from different angles, ranging from text understanding to generation (Alhuzali and Ananiadou, 2021; Schmidt et al., 2021; Gao et al., 2021; Wang et al., 2022). Its efforts have spurred the creation of many resources for processing affect in text (Bostan and Klinger, 2018), mostly based on psychological theories that treat emotions as discrete labels, or vectors in a space defined by the dimensions of valence, arousal and dominance (Ekman, 1992; Plutchik, 2001; Russell, 2012), allowing machine learning (ML) approaches to model emotions in the same way (Buechel and Hahn, 2018; Demszky et al., 2020; Hipson and Mohamad, 2021). Current research disregards, however, more complex mechanisms that govern the emer-

Strategy	Example
Attack	Every word you said just proves how utterly clueless you really are. I won't tolerate your ideas anymore.
Contact	Allow me to disagree. I feel like your ideas might have fatal repercussions.

Table 1: Texts expressing different coping strategies.

gence and subsequent effect of emotions on behavior, modeled in psychology, e.g., as coping processes (Lazarus, 1991a; Scherer, 1982). Essentially, emotions can be deemed episodes in which people *cope* with an event, and which affect their behavior through action tendencies (Lazarus, 1991b). Roseman (2013) posits that there are four possible ways in which this happens, corresponding to four coping strategies by which humans *attack* the emotion stimulus, *contact* it, *distance* themselves from it, or *reject* it, contextual to specific emotions.

Coping theories have visibility in research on virtual agents and robotics (Spekman et al., 2018, i.a.), due to their ability to model emotional phenomena at the behavioral level, but their potential for understanding emotions in language (as a specific type of behavior) has not been fully explored so far. In this paper, we propose to treat textual emotions in terms of coping. As computational research moves towards the consideration of emotions as episodes fundamentally centered on events (Klinger, 2023), automatic systems are required to understand how such events are coped with when individuals feel emotions and express them verbally: they urge us to question how coping strategies, an essential building block of communication, realize in language, and if they can be captured by ML systems.

We hypothesize that these strategies emerge in text just like the emotions (anger, hope, etc.) that set them into place. Intuitively, a text expressing anger should also convey the intention of attacking

a stimulus, and one associated to hope might verbalize a contacting-type of coping (see Table 1 for some examples). Our study sets out to verify this assumption: we construct COPING (COPING emotion strategies), the first corpus focused on coping information, and we investigate the feasibility of the task of coping strategy detection.

We approach this task using dialogues that depict conflictual situations about topics of actual relevance, in which one interlocutor utters a controversial statement threatening one’s identity, and the other person fights against it while implementing a coping strategy. In ecological settings (e.g., on online platforms), finding interactions of this type annotated with coping information is hard. Hence, we gather them via crowdsourcing, building COPING with an experimental design for text generation and annotation based on role-playing. Role-play provides a methodologically founded way to elicit behavioral and linguistic responses by asking individuals to put themselves in the shoes of a fictional (yet credible) character, with whom they do not necessarily share concerns and values. In our case, people impersonate a character that portrays one of Roseman’s coping strategies, and produce text to disagree with an imaginary interlocutor.

Using COPING, we answer the following research questions. (RQ1) Can coping strategies be elicited in language and subsequently annotated? With a quantitative and qualitative analysis of the corpus, we show that this is the case – with some caveats deriving from our data collection procedure. (RQ2) Can coping strategies be identified automatically? To answer this question, we fine-tune models for the classification of coping strategies on COPING, and compare them to humans and four prompt-based Large Language Models (LLMs) performing the same task in a zero-shot setup.

In sum, our contribution is threefold. (1) We take an initial step to study the four coping categories that encompass the labels typically used in computational emotion analysis, driven by Roseman’s taxonomy of 16 emotions based on attack, contact, distance and reject (see Table 2). (2) We release a highly-structured corpus, built with a novel strategy for the collection of emotion-centered information. (3) We introduce and benchmark the task of coping strategy identification. Experimental results support its viability.

Our corpus and code are available at <https://tinyurl.com/y6b47tsc> during review.

Coping Strategy	Emotions	Function
Attack	Anger, frustration, guilt	Move against stimuli
Contact	Joy, hope, love, relief, pride	Increase contact and interaction with stimuli
Distance	Dislike, distress, fear, regret, sadness	Decrease contact and interaction with stimuli
Reject	Contempt, disgust, shame	Move stimuli away

Table 2: The four emotion groups with their coping strategies and function, as in Roseman (2013).

2 Related Work

Below, we summarize the concepts of coping and emotions, and related NLP research.

2.1 Coping Strategies and Emotions

According to well-established psychological theories, emotions emerge as one evaluates, or appraises, salient situations (Scherer et al., 2001). For example, if someone receives a critical review, they might appraise it as a sign of failure and feel anxious, or appraise it as constructive and feel determined to improve. Within this theoretical framework, Roseman (2013) links emotions to coping, a process useful for “responding to situations of crisis and opportunity” (p.143).

More precisely, emotions are instances of the general strategies of *attack*, *contact*, *distance*, and *reject* (as in Table 2), that serve to seek more or less of a stimulus. For example, emotions like anger and frustration correspond to the *attack* strategy by which people engage with a source of distress; hope and joy pertain to the *contact* strategy, which increases one’s interaction with the stimulus; fear and sadness belong to the *distance* strategy, that minimizes the source of distress; disgust and shame share a *reject* strategy to eliminate the stimulus.

Roseman’s theory is accompanied by others on the same matter, such as that of Lazarus and Folkman (1984), which sees coping as two simultaneous stages: (1) a person evaluates the motivational relevance and congruence of a situation in respect to their own goals: “How relevant

and (in)consistent is the situation with my own needs/goals?” (Smith and Kirby, 2009); and (2) the situation is evaluated in terms of coping potential: “Which resources and options do I have for coping?” (Smith and Kirby, 2009). This theory further distinguishes between problem-focused and emotion-focused coping. Problem-focused strategies are directed at the stimulus (e.g., decreasing environmental pressures) or at the self (e.g., altering level of aspiration) (Kahn et al., 1964), while emotion-focused coping involves cognitive processes that reduce emotional distress (e.g., avoidance or acceptance) or increase it to mobilize for action (Lazarus and Folkman, 1984). Although highly influential, the distinction between problem- and emotion-focused coping was criticized in later research for being non-exclusive and non-exhaustive (Skinner et al., 2003). We therefore follow the theory of Roseman (2013).

2.2 Psychological Theories and NLP

NLP works mostly focus on the use of discrete emotion labels to extrapolate from text, borrowed from theories such as Ekman’s (1992) and Plutchik’s (2001). There have also been successful attempts at identifying the underlying components of emotions, in terms of valence, dominance and arousal (Mohammad, 2018; Park et al., 2021), following dimensional models of affect (e.g., Russell, 2012). More recently, a line of research has considered the entanglement between emotions and their eliciting events to carry out a discussion and the modeling of appraisals (Yeo and Jaidka, 2023; Zhan et al., 2023). Our work relates to the latter direction.

We also align with contributions that tap on psychological theories of *re-appraisal* (Uusberg et al., 2019), which document how people re-interpret a situation to alter its emotional impact (e.g., “When I realized I could listen to my podcast, I stopped feeling frustrated about the traffic jam”). For instance, based on reappraisal, Li et al. (2024) and Ziemis et al. (2022) reframe texts that depict negative situations. In contrast to them, we use texts that induce distress in annotators, as they describe situations that are threatening for one’s identity, and that activate coping (similar to Saha et al. (2019) and Choi et al. (2020)). We therefore focus on coping instead of reappraisal, to study if and how the four Roseman’s strategies emerge in text produced in response to adverse events.

3 Data Creation Methodology

Textual data associated with coping strategies is hard to obtain. Ideally, since coping is a response to an appraised situation, both the situation itself and the textually expressed coping strategy should be known – but the former may be external to the text and impossible to track. Moreover, coping strategies rarely appear as naturally-occurring labels of texts (e.g., as hashtags assigned by writers), and involve complex concepts (i.e., the psychological sense of *attack*, *contact*, *distance* and *reject*) which might be unfamiliar to laypeople. We solve both issues by resorting to a role-playing approach inspired by the paradigm of improvisational theater (Baumer and Magerko, 2009) and supported by research on character identification (Giovannelli, 2009; Battaglino et al., 2014): participants engage in a make-pretend activity where they write texts (instead of annotating existing ones) while (purposefully) instantiating a given coping strategy.¹ In practice, we approximate the collection of data from first-hand experiencers of coping.

We apply this approach as in Figure 1. We start by gathering the building blocks of the role-playing study (Section 3.1), namely, definitions for the four strategies and texts depicting scenarios in which these might be triggered. Based on such definitions and scenarios, we stimulate the production of coping-expressing utterances (Section 3.2).

3.1 Building Blocks of Role-playing

Good definitions of the strategies under considerations retain Roseman’s insights and minimize his technical jargon. We thus formulate each strategy s as the profile of a fictional character called X , who has all idiosyncrasies of that s . We use these descriptions² to crowdsource the task of coping strategy recognition, where people infer a coping strategy from a given description of s . This labeling task t can be conceptualized as estimating $p(s \mid \text{description}(s))$, and it is conducted with the rationale that the annotators’ performance will signal the description quality: the more a description is mapped to the correct s , the clearer it is.

As for the scenarios, we start from the idea that coping occurs in moments of crisis and opportunity, and we identify 5 topics of current social relevance which might spur emotions and per-

¹Workers are recruited on Prolific.

²Descriptions are in Appendix A, with guidelines (Table 8) and results (Figure 3 and 4, Table 9 and 10).

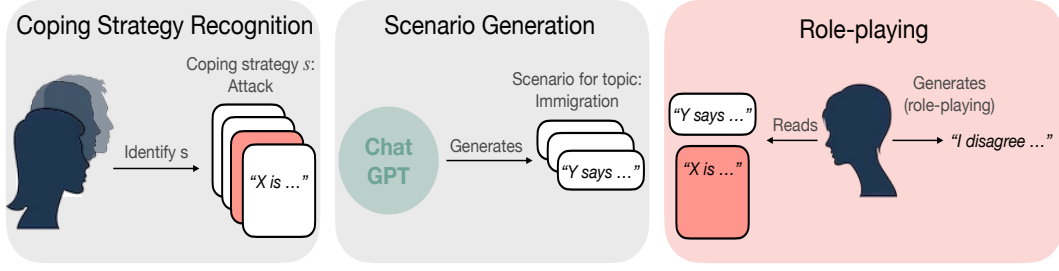


Figure 1: Crowdsourcing methodology. Grey boxes: collecting the building blocks for role-playing. Annotators identify the coping strategy s (in the figure, attack) in the description of character X ; separately, ChatGPT-4 produces scenarios for different topics (in the figure, immigration). Pink box: role-playing. A character description and a scenario on a given topic are presented to annotators as textual prompts to induce coping-related text generation.

sonal stances, i.e., abortion, immigration, racism, LGBTQ+ rights, and drugs. We generate 3 controversial scenarios (i.e., fictional dialogical settings, in which an interlocutor puts forward a contentious argument) about each of them. An example coping definition and scenario are in Table 3.

Coping Strategy Recognition. Our goal is to observe if crowdworkers associate X with the correct coping strategy: presented with a description of X , the annotators pick one of four possible strategies. We conduct this task in four rounds, each time modifying both the descriptions of X , and the set of possible answers presented to the annotators.³ From the candidate descriptions collected from all rounds, we select the four that elicit highest annotation accuracy (attack and contact: 100% participants recognized the correct strategies, distance: 92%, reject: 52%) to write the guidelines for the role-playing task of Section 3.2.

Scenario Generation. The scenarios are gathered via ChatGPT-4 to mitigate our personal and cultural stances.⁴ We ask the LLM to produce three descriptions of every-day situations (e.g., meet-ups in a bar), in which participants X and Y interact, and Y utters a strong opinion on a sensitive topic. ChatGPT-4 is required to specify the power status between X and Y (i.e., they are either two peers, or have a power imbalance), as we assumed that the way people respond to Y could depend on what relationship they hold with such interlocutor.

³Initially, the four answer options comprised the coping strategy labels (e.g., *attack*); later, we opted for one-line summaries of X 's profile – e.g., *X directly approaches problems with a potentially conflicting attitude (i.e., X is an attacking-kind of person)*.

⁴All prompts for ChatGPT-4 are in Appendix B, Table 11. The generated scenarios for each topic can be found in Table 12.

3.2 Role-Playing

We obtain texts associated with coping strategies by crowdsourcing task $t' = p(\text{generation} \mid t)$: participants⁵ generate text based on their understanding of a description of s . Their assignment is not only to form a mental representation of X and write a text: they must impersonate X and write as X would, in such a way that the resulting text discloses X 's personal features. Since X represents a coping strategy, we assign the texts so collected to the corresponding s .

In practice, we collect the texts with 60 questionnaires, each being a combination of a description of X and a scenario on a given topic. All questionnaires ask participants to map the description of X to a short summary of X 's profile (as in the previous task t). Once they have familiarized themselves with X , they read the scenario (which includes the hostile interlocutor Y), and they: (i) generate a reply that opposes Y 's statement, by pretending they were X (cf. Table 3); (ii) describe X 's non-verbal reaction; (ii) rate the emotional response of X according to Roseman's taxonomy of emotions, on a scale from 1 to 5. (iv) Finally, participants are asked if they themselves would react verbally and non-verbally like X . If their answer is negative, they repeat the first three steps from their own perspective instead of X 's.⁶

The writers' responses could be influenced not only by their role-playing abilities, but also by their own involvement in the topic under consideration. Hence, to elicit bolder opinions, potentially resulting in clearer coping strategy expressions, we select participants with specific demographic features for each topic (e.g., people from the LGBTQ+ community for the LGBTQ+ rights topic).

⁵These workers do not overlap with those in Section 3.1.

⁶Details on questionnaires and participants in Appendix C.

Definition	This character comes across as a calm, understanding, and very approachable person. For X, communication serves to unite people. It is an opportunity to exchange opinions, acknowledging the diversity of perspectives among individuals. When problems or unpleasant situations arise, this character responds with a constructive attitude. X expresses ideas with confidence, trying to solve problems in a respectful manner. X can effectively engage in discussions also with people having contrasting opinions.
Scenario (topic: racism)	During a university class discussion on historical racial events, Y confidently states, “People keep talking about systemic racism, but I believe that’s just an excuse for those who don’t want to work hard. If you look around, everyone has the same opportunities today.”
Generated text	I understand your point Y, but it is not the case for everyone. Our group is a select handful of people who have been brought up this way.

Table 3: Example description of X (i.e., definition for the coping strategy contact), with a scenario for the topic racism, and a text generated by a participant presented with the former, during the role-playing phase.

4 The COPING Corpus

COPING contains 1,200 data points, balanced among coping strategies, topics, and scenarios. On average, the length of X ’s answer to Y is 34.8 tokens⁷, that of texts about X ’s behaviors is 23.6. We now analyze the corpus to investigate if the four descriptions of X (such as that in Table 3) allow for a correct understanding of coping strategies. By that, we answer (RQ1) **Can coping strategies be elicited in language and subsequently annotated?** We investigate coping understanding from different angles. First (Section 4.1), irrespective of how this understanding reflects in text – to separate such an understanding from people’s ability to write utterances while role-playing. Then (Section 4.2), by questioning how well texts signal the presence of specific coping strategies.

4.1 Coping from the Perspective of Writers

Our work adheres to Roseman’s framework, but does so in an “artificial” experimental setup. We therefore test if the emotion judgments provided by participants are theoretically expected. If participants correctly impersonated X (i.e., a given s), that has to reflect in a distribution of emotions similar to Roseman’s (cf. Table 2) – attack, e.g., should be associated to a high level of anger, frustration, and guilt. Further, we observe how such emotions are different from the annotators’ own emotions.

Emotions of Character X . Figure 2 illustrates the distribution of emotions per coping strategy. Different coping strategies are characterized by different emotion configurations: the mapping between Roseman taxonomy and the judgment of the crowdworkers is correct for most emotions (e.g., frustration, hope, dislike, contempt for attack, contact, distance, and reject respectively). There are,

however, exceptions. For example, crowdworkers failed to associate guilt to attack and shame to reject, and love and relief to contact. In addition, they assigned high values to, e.g., frustration in contact and anger in distance: while theoretically imprecise, such scores are likely an effect of the scenarios that crowdworkers were presented with, which faced them with conflictual situations (i.e., likely inducing negative emotions).

We corroborate these insights through correlations between emotions and coping strategies. We compute the Pairwise Cramér’s V association (Cramér, 1999) between the two variables, and deem them correlated only when $V > .3$ (i.e., moderate to strong correlations). To decide which coping strategy an emotion is most strongly associated with, we consider its average score, and assign it to the coping strategy where that average is the highest – e.g., anger correlates with attack because $V > .3$ and the average respondents’ rating is highest on that strategy (4.14) than any other. Results are in Table 4, where the emotion patterns underlying each strategy suggest that the crowdworkers achieved a sufficient level of understanding of character X . Six emotions (marked in bold) prove coherent with Roseman’s framework. For instance, frustration correlates with attack, hope with contact, fear with distance.

Vice versa, lacks of correlations for joy, pride, regret, relief, and sadness indicate that these emotions were not well identified by the participants. Rather than a negative outcome, we take this observation as an indicator of the success of role-playing (making positive emotions like joy and pride unlikely). Unsurprisingly, reject turns out uncorrelated to any emotion, since those documented in Roseman’s account (contempt, disgust, shame) are (incorrectly) more linked to other coping strategies. This outcome points not only at the difficulty for

⁷Tokenization performed with spacy en_core_web_sm.

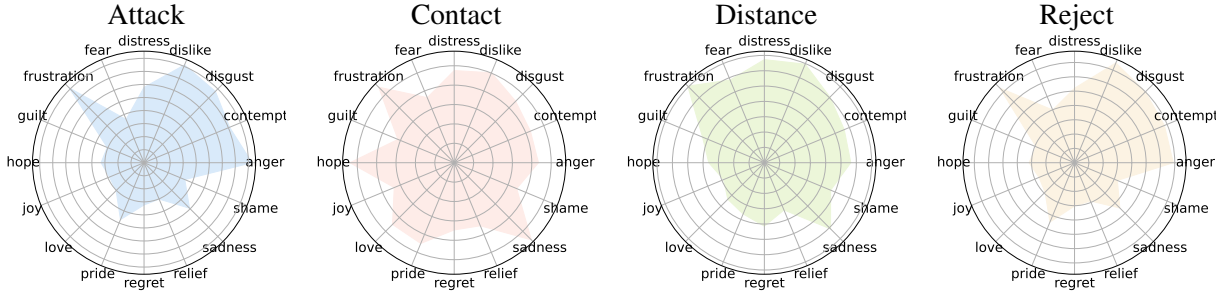


Figure 2: Emotion averages of character X for each coping strategy.

Strategy	Emotion
attack	anger frustration contempt disgust dislike
contact	hope love
distance	distress fear shame guilt
reject	–

Table 4: Coping strategies and correlated emotions (Cramer’s $V > .3$). Emotions included in Roseman’s framework for a given strategy are marked in bold.

respondents to identify reject (cf. Section 3.1), but also at the fact that such emotions are interwoven with others. For instance, in Table 4, shame falls under the same coping strategy as guilt – both are indeed related to self-forgiveness (Leach, 2017); disgust is, like anger, a response to a moral violation (Molho et al., 2017) and part of the so called CAD triad (Contempt, Anger and Disgust) of moral-based emotions according to Rozin et al. (1999) – the three are often elicited in combination in hostile situations (Matsumoto et al., 2016), which might explain why they are all correlated with attack.

Therefore, even if not perfectly aligned with Roseman’s taxonomy, the workers’ understanding of coping appears meaningful, based on both theoretically-established links between emotions and our experimental setup for data collection.

Effect of Role-Playing. We now observe if the participants disentangled their own emotions from X ’s, further corroborating their understanding of coping. We focus on the answers of the 571 respondents (47.5% of the crowdworkers) who claimed that they would not react as X , and we test this self-assessment by performing a Wilcoxon (1945) signed-rank test between emotional scores provided while role-playing X and scores provided from their own perspective (e.g., between

the scores of X ’s anger and the participant’s). We find that claiming a for a different personal reaction corresponds indeed to different emotion reactions: except for pride (p-value: .06), all pairs of values for the same emotion are significantly different.⁸ In sum, X ’s and the annotators’ emotions can be clearly separated.

4.2 Coping from the Perspective of Readers

Focusing only on the textual responses gathered with role-playing, we now reverse the direction of inquiry, and question if the texts express the intended coping strategies according to readers.

Inter-Annotator Agreement. We conduct an in-house annotation study to see if said strategies are successfully recognized by external people. We randomly select 100 texts and have 3 annotators decide if and what coping strategy emerges from each of them. Inter-annotator agreement computed with Fleiss’ kappa (1971) shows that the annotators’ judgments were not consistently similar ($\kappa = .34$). Annotators achieved the highest agreement on the strategy of contact (.6) and the lowest on reject (.04). On 42 texts, they provided unanimous judgments. However, on only 28 of these items were the correct prompting labels recognized (mostly, attack and contact). Surprisingly, a majority vote was reached on only a few more texts (45); that vote matches the correct label for 18 of those. This finding hints at the fact that the intended coping strategies are recognized when they are as clear as to put many people in agreement.

Differences in Linguistic Patterns. To better understand these results, we look for significant forms of variation between texts pertaining to different coping strategies. For each strategy, we split the

⁸A comparison between X ’s emotions and the respondents’ self-assigned emotions is in Appendix D.

Strategy	Tokens
attack	you, wrong, stupid, dare, racist
contact	understand, help, feel, while, however
distance	I, subject, sure, but, opinion
reject	do, not, negative, want, attention

Table 5: The five tokens most strongly associated with each strategy.

corpus in two parts: the first with all the texts related to the given strategy, and the second with all the remaining texts. We then compare the distribution of tokens in the two sets, by computing the Jensen-Shannon divergence (Menéndez et al., 1997) for each token.

Table 5 lists the five tokens most strongly associated with each strategy. A set of tokens appears highly coherent with attack (e.g., wrong, dare) and contact (e.g., understand, feel). A manual qualitative analysis of the texts corroborates this finding. Contact might be relatively easy to recognize thanks to politeness markers (e.g., “*Allow me to disagree*”), while the detection of attack corresponds to opposite linguistic strategies, such as pressing questions (“*How are they creating chaos by just existing?*”), direct confrontations (“*Only a fool wouldn’t not see the obvious*”), and expressions of how strongly *X* feels about their opinion or the interlocutor’s (e.g., “*I could not disagree more*”, “*you are completely wrong*”).

Specific linguistic patterns are less evident for distance and reject, associated with tokens that vaguely express them. The difference with the other two coping strategies might be explained referring to Roseman’s theory. Attack and contact require interaction with the emotion stimulus; distance and reject disengage from it. Role-playing disengagement might be more challenging, because it requires participants to react to a stimulus while detaching themselves from it. Ultimately, these observations point out that some coping strategies are easier to be imbued (and recognized) in text, at least when they are purposefully produced.

5 Coping Strategies Classification

(RQ2) Can coping strategies be identified automatically? We answer our second research question using fine-tuned and prompt-based language models for coping strategy detection.

Models. We fine-tune three Small Language Models (SLMs) on the COPING texts: BERT (Devlin et al., 2019), DistilBERT (Sanh et al., 2019), and RoBERTa (Liu et al., 2019). The fine-tuning is performed on three separate sets of texts: *X*’s answers, *X*’s behaviors, and their concatenation. After splitting the corpus in a train and a test set (90% and 10% of the data), we fine-tune each model over three random train/validation splits and train it for 10 epochs with a batch size of 10.⁹ As for the prompt-based models, we implement zero-shot classifiers with four open-source LLMs, i.e., phi-2, mistral-2, nous-hermes-2, dolphin-2, reproducing the design used with SLMs (answer, behavior, and their concatenation) on the same test data.¹⁰

Results. The classification of coping strategies turns out challenging, both for the SLMs and the prompt-based ones. As shown in Table 6, which reports the best F_1 score obtained by the classifiers on the test set, all results are below .55.

We observe differences for different types of input texts. DistilBERT has the best classification performance when fed with answers ($F_1 = .523$), while RoBERTa performs better in the classification of *X*’s behaviors, with a score of .537. The combination of replies and reactions does not lead to a better performance: the best model is RoBERTa with an F_1 score of .512, which is lower than the ones obtained for replies and reactions separately. The zero-shot classifiers are on par with the others. Once more, the outputs based on behavior alone surpass the two other setups. Here, the difference between Answer and Behavior is much more noticeable (12 points in F_1 for dolphin-2, ≈ 14 F_1 points for nous-hermes).

Zooming in topic-wise results (Table 7), we see that both answers (.623 F_1 score) and behaviors (.630) in scenarios related to abortion are better recognized by the fine-tuned DistilBERT (answers) and RoBERTa (behaviors). The lowest F_1 score of answers are obtained on immigration-related scenarios (.365), while the worst classification of reactions on racism-related scenarios (.457). We also notice that not all coping strategies are equally difficult to detect (cf. Table 7). The easiest is contact, classified with an F-score of .647 using *X*’s answers and .617 using behaviors. Conversely, the classification of reject results in the lowest F-

⁹Each training has been performed on a NVIDIA RTX-3060 Ti and lasted 25 minutes on average.

¹⁰Appendix E details models, prompts, and full results.

	Answer	Behavior	Answ.+Behav.
BERT	.401	.395	.457
DistilBERT	.523	.443	.507
RoBERTa	.477	.537	.512
phi2	.356	.391	.322
mistral2	.392	.474	.538
dolphin2	.422	.550	.520
nous-hermes	.387	.526	.515

Table 6: Top: F_1 scores of the SLMs ran on X ’s answers (Answer), X ’s behavior (Behavior), and a concatenation of both (Answ.+Behav.). Bottom: F_1 scores obtained by feeding the four LLMs (rows) with a concatenation of instructions and input text (Answer, Behavior, both).

		Answer	Behavior
Topics	Abortion	.623	.630
	Drugs	.513	.488
	Immigration	.365	.483
	LGBTQ+	.508	.619
	Racism	.570	.457
Labels	Attack	.500	.545
	Contact	.647	.617
	Distance	.539	.560
	Reject	.408	.428

Table 7: Classification results broken down by topic and coping strategy. Answers have been classified with DistilBERT, behaviors with RoBERTa.

scores: .408 for answers and .428 for behaviors. This aligns with the human recognition of coping strategies, with contact and reject defining the best and worst recognized labels using character X ’s answers (cf. Section 4.2).

Put together, these numbers suggest that X ’s answers provide coping signal, but the texts about X ’s behavior are key to capture the “seeking more” or “seeking less” of an event, central to the understanding of coping and its link with behavior.

6 Discussion and Conclusion

This paper follows a recent wave of studies (Cortal et al., 2023; Liu and Jaidka, 2023, i.a.) that initiated a dialogue between computational emotion analysis and theoretical disciplines (like psychology), to broaden its tools and push forward its potential to shed light on how emotions work in language. We focused on the equivalence between emotions and coping strategies, never-before explored in NLP. Our goal was to understand if emotion-bearing texts allow to infer affective information that goes beyond discrete emotion experiences (as captured

by emotion labels), and extends to their core motives of “seeking more or less” of a stimulus.

We built a corpus for coping detection via role-playing, to overcome the lack of available textual data annotated with coping information. Role-playing facilitated our attempt to communicate abstract constructs to laypeople, and to gain control over many variables (e.g., all people had the goal of realizing a coping strategy, in response to the same controversial statements, on a small set of topics) but at the same time it introduced some artifacts. Notably, the expression (and identification) of coping strongly relied on our ability to communicate them in X ’s descriptions, as well as people’s ability to understand such descriptions, to engage in role-playing and produce good-quality, coping-expressing texts.

Therefore, both the quality of the texts and the classification results are to be understood by considering the challenges introduced by our data collection procedure: humans and classifiers alike provided evidence that the emergence of coping from text is not as straightforward as we initially hypothesized. However, this conclusion is not definitive, as one could still expect that, were the texts produced for counter-arguing purposes (like ours) but spontaneously (unlike ours), the attack, contact, distance, or reject mechanisms might be clearer. In fact, the possible recognition of coping, and the idea that these strategies lend themselves for annotation is corroborated by our analysis of their link to emotions – with meaningful patterns as found in Roseman’s literature. This hints at the potential use of coping for future tasks.

Identifying coping strategies computationally could indeed benefit emotion analysis in multiple ways, as it can be instrumental for different downstream objectives, for instance to understand argumentative strategies, revealing how people react to verbal conflicts. Leveraging coping strategies is also valuable to establish connections with other theoretical frameworks. One can identify coping strategies in a multi-task setup to better identify the emotions of a text (e.g., based on Ekman’s theory), or to explore their link with appraisal theories in event-centered emotion analysis (Klinger, 2023), to explain how events (e.g., those described in our scenarios) are evaluated or re-appraised (cf. Li et al., 2024; Morris et al., 2015). Such integrated approaches promise a comprehensive understanding of emotions, still ripe for further investigation.

Ethical Statement

The role-playing study has been approved by an ethical committee collaborating with our affiliations. Crowdworkers were rewarded with a fee that corresponds to 9£ *per* hour for their participation.

Since the texts in COPING are based on conflictual dialogues, they contain potentially sensitive material that might depict real-life experiences of some individuals. All crowdworkers were warned beforehand, and were informed about the type of debate they would have entered by taking part in the study. Their IDs in the corpus have been encrypted to prevent identification.

Both the fictional interlocutor (whose utterances were produced by ChatGPT) and *X* (impersonated by people) express standpoints that can have political significance. We remain neutral with respect to both sides.

Data and code are released under Apache 2.0 license

Limitations

While we deemed COPING appropriate to conduct a proof-of-concept study, we acknowledge that its size is suboptimal to extrapolate features of coping strategies, and train classification models.

Its construction relied on several preliminary rounds of crowdsourcing which corroborated the difficulty of our tasks for laypeople. We simplified it via role-playing, but this strategy required us to embrace some simplifying assumptions. First and foremost, the texts were not generated while experiencing the emotion(s) necessary to implement a coping strategy. The conflictual situations depicted in the scenarios had the potential to stir an emotion, but we did not measure their success in this sense, and we recognize that imagining to feel an emotion is not the same (in terms of cognitive and physiological processes) as actually feeling one.

To mitigate this problem, we assigned participants to the topics they might feel more related to by opening the studies to crowdworkers with specific demographics (e.g., women for the abortion topic). This decision, too was based on simplifying assumptions, namely, that demographics approximates one's experiences and involvement in the given topic, and that the understanding of laypeople would reflect on their ability to generate texts.

Overall, our setup does not necessarily correspond to one that psychological research on coping would adopt: albeit useful, coping strategies still

require us to reflect on how to more seamlessly connect emotion-based NLP and the disciplines from which we borrow theoretical constructs.

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A Describing Coping Strategies

The formulation of coping strategies described in Section 3.1 was based on 4 crowdsourcing rounds. Round 1, 3, and 4 involved 100 participants each (25 per coping strategy). In round 2, we only focused on attack and reject, with the goal to maximize their difference (25 people participated for each of these strategies). We selected participants on the basis of first and primary language (i.e., English), approval rate on past Prolific studies (between 90% and 100%). Participants could only take our study once (i.e., for no more than one coping strategy, in no more than one round). The total cost of this phase was 210£.

Each round comprised 4 questionnaires, one per coping strategy. All questionnaires included the same questions, one for the identification of coping strategies and two others, to (1) rate how much of a given list of emotions are likely felt by character *X*, when confronted with an unpleasant situation, and (2) associate (the Big-Five) personality traits to *X*. The emotion list included those analyzed by Roseman (2013): anger, contempt, disgust, dislike, distress, fear, frustration, guilt, hope, joy, love, pride, regret, relief, sadness, shame. Each of them was

rated on a 4-item scale (i.e., from *The emotion does not apply* to *The emotion very much applies*). We additionally collected personality traits because the coping understanding task presents coping strategies in terms of personalities.

Emotion information served to observe if the annotators’ associations of emotions to coping strategies mapped to the original taxonomy proposed by Roseman, in which each strategy underlies a handful of emotions (similar to what we did in the role-playing data, in Section 4.1). Instead, information about the traits of *X* served to analyze if *X*’s descriptions led people to form similar understandings of its persona, in terms of openness to experience, conscientiousness, extroversion, agreeableness and emotional stability.

We detail tasks and corresponding results below.

A.1 Annotating Coping Descriptions

The description of character *X* in a coping-specific questionnaire changed from round to round. Figure 3 reports the confusion matrices for the successive rounds of this crowdsourcing step, indicating the number of annotators who correctly associated *X* description to the corresponding coping strategy (diagonal cells) and how many made incorrect associations (off-diagonal numbers).

The descriptions entering the role-playing guidelines are those found in the last round (see Table 8). Thanks to such descriptions, the annotators reached perfect understanding of attack and contact, 92% of the annotators correctly chose the distance strategy, and 52% made the correct inference for reject.

A.2 Additional Tasks

Besides associating character’s *X* description to a one-liner summary of the corresponding coping strategy, the annotators judged emotions and personality traits.

Emotions. In order to better assess the performance the annotators, we investigated how similar the emotion understandings of wrong and correct annotators are. More precisely, we observe:

1. if the annotators rated the emotions similarly, despite their coping choices being different (if so, we can move on to the generation phase);
2. to what extent they captured the “correct” emotions (i.e., those assigned by Roseman to each coping strategy) – once more, this

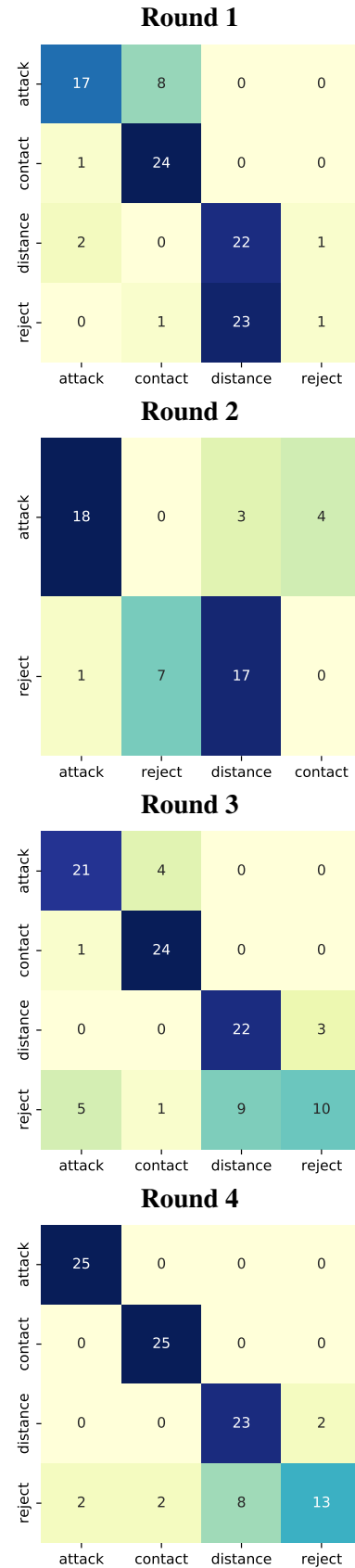


Figure 3: Confusion matrices for the pre-test rounds. Rows are the annotated labels. Columns are original labels. Numbers correspond to the annotators (in total, 25 for each strategy).

Strategy: Attack Description: X is a very opinionated person, who comes across as aggressive and combative. When problems or unpleasant situations arise, X responds with a hardened resolve. X gives their all to overcome obstacles, and never shies away from a conflict. In fact, this character faces negative situations fiercely and with intense energy, taking action personally in order to right the wrong. For this reason, one can often see X all tense and pumped up. Short summary: X rights the wrong in an aggressive manner, like in the picture below (X is shown in blue).	
Strategy: Contact Description: This character comes across as a calm, understanding, and very approachable person. For X, communication serves to unite people. It is an opportunity to exchange opinions, acknowledging the diversity of perspectives among individuals. When problems or unpleasant situations arise, this character responds with a constructive attitude. X expresses ideas with confidence, trying to solve problems in a respectful manner. X can effectively engage in discussions also with people having contrasting opinions. Short summary: X rights the wrong with an open mindset, like in the picture below (X is shown in blue).	
Strategy: Distance Description: In life, X has managed to create a peaceful mental space as a protection from the outside world. This character always seeks emotional tranquillity, and finds solace in self-reflection. When problems or unpleasant situations arise, X feels powerless, not too capable to change the external world. So here's X's defense strategy against all negativity: staying away from stressful things, events, people or thoughts, to minimize their influence. At the cost of coming across as an avoidant person, X does not engage in conflicts and uncomfortable situations. Short summary: X seeks shelter from unpleasant situations in a quiet inner space, like in the picture below (X is shown in blue).	
Strategy: Reject Description: This person has a judgmental attitude towards others. X easily disagrees with people, and is often displeased with their behaviors and beliefs. Here's X's motto in life: do not let anybody interfere with your well-being. This character keeps negativity away at all costs. It's not that X hides from problems and tries to find solace in a peaceful mental space, it's more of a resolute attempt to refuse unpleasant situations. It's also not a lack of courage; in fact, X expresses opinions with confidence, and precisely by this, X manages to dismiss conflicts with other people. Short summary: X's approach to life is a firm refusal of negativity, like in the picture below (X is shown in blue).	

Table 8: Descriptions of *X* used in the final round of coping recognition and in the role-playing crowdsourcing phase. Each coping strategy is presented as a description of character *X*, and in a more concise version (short summary), and with the reported picture.

would be a sign that we can proceed with role-playing phase of crowdsourcing.

using a 7-point Likert scale ranging from “disagree strongly” to “agree strongly”.

1033
1034

The answer to point 1. emerges from Figure 4, Table 9 and Table 10, which contain results from the last round.

Figure 4 shows the average values assigned to each emotion, for each coping strategy. The values obtained from the annotators who correctly understood the coping strategy are side-by-side the values computed on the answers of the annotators who did not get the correct strategy.

While the figure tells us “how much” the annotators rated an emotion, Table 9 and 10 summarize this information in a binary way (did the annotators assign a given emotion to a coping strategy?). The symbol ✓ means that a given emotion was recognized. Note that emotions were rated on a scale from 1 to 5. To decide if an emotion counts as correctly recognized, we observe if it has an average rating of >2. The numbers next to ✓ correspond to the ranking of the emotion (e.g., 1 means that the emotion for that coping strategy had the highest average rating).

From both Figure 4 and the tables, we see that the emotions captured by the two groups of annotators overlap for all coping strategies.

Concerning point 2., we answer it by looking at Table 9 and Table 10. Both report the annotators’ judgments next to the correct emotions indicated by Roseman. There, we see that for none of the coping strategies the annotators understood all relevant emotions. However, the correct emotions that the crowdworkers captured have the highest ranks (e.g., focusing on attack, anger and frustration have the highest ratings, for both groups of annotators).

Personality Traits. Crowdworkers estimated how much the 10 items in the questionnaire of Gosling et al. (2003) might apply to character X. The questionnaire evaluated openness to experience by contrasting “open to new experiences and complex” against “conventional and uncreative”; conscientiousness through the trait “dependable and self-disciplined” versus “disorganized and careless”; extraversion by comparing “extraverted and enthusiastic” with “reserved and quiet”; agreeableness by distinguishing between “sympathetic and warm” and “critical and quarrelsome”; and emotional stability by assessing “calm and emotionally stable” against “anxious and easily upset”. Respondents rated X on these dichotomous descriptors

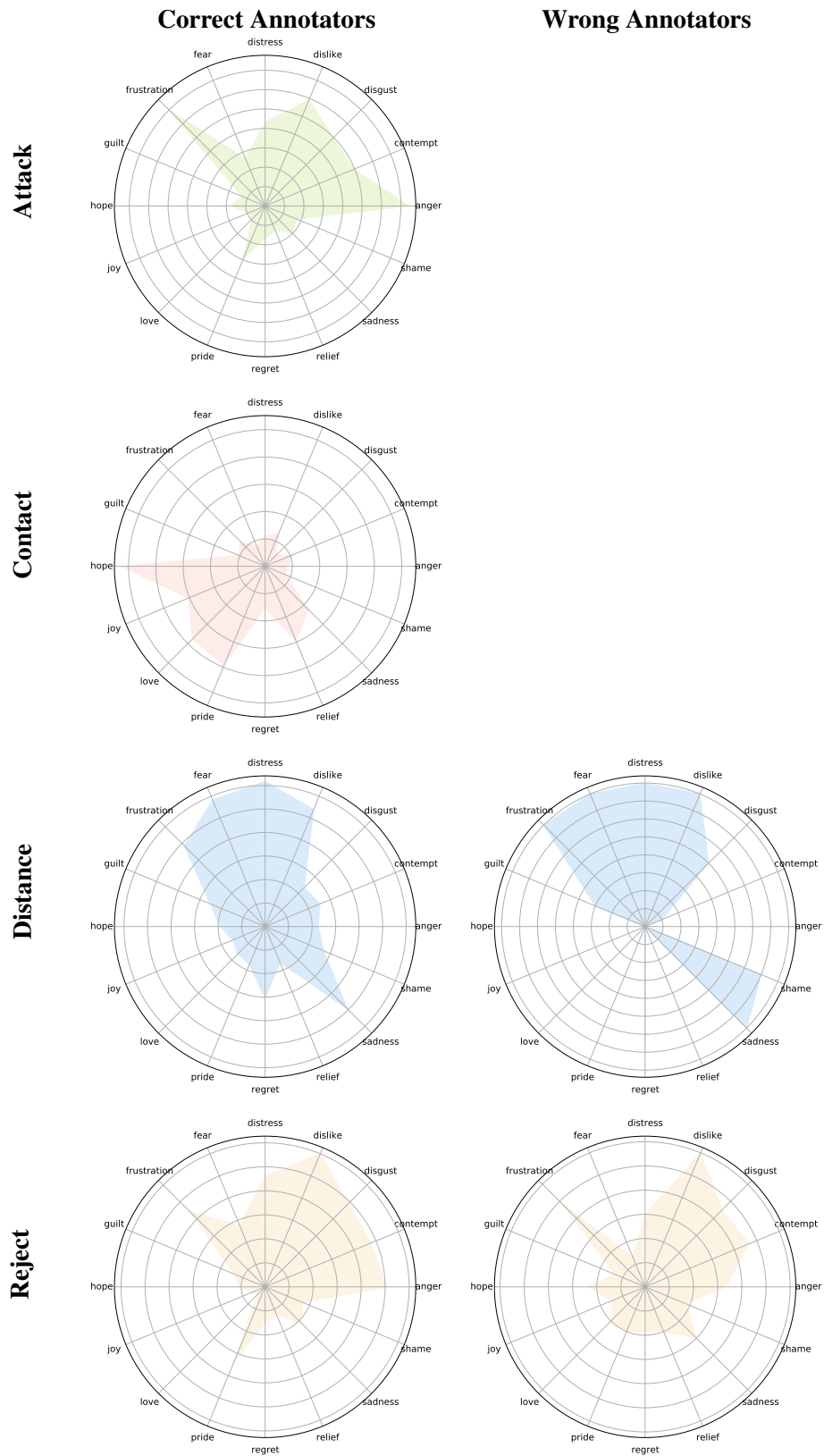


Figure 4: Plots of the emotion answers (averages) for the last round of crowdsourcing coping strategy recognition, divided by annotators who were right about the coping strategy (Correct Annotators) and those who did not correctly recover the strategy from the description (Wrong Annotators).

Emotions	Attack		Contact	
	Original	Correct Annotat.	Original	Correct Annotat.
Anger	✓	✓, 1		
Contempt		✓, 5		
Disgust		✓, 4		
Dislike		✓, 3		
Distress		✓, 6		
Fear				
Frustration	✓	✓, 2		
Guilt	✓			
Hope			✓	✓, 1
Joy			✓	
Love			✓	
Pride			✓	
Regret				
Relief			✓	
Sadness				
Shame				
Surprise				

Table 9: Emotions for the strategy Attack and Contact as indicated by [Roseman \(2013\)](#) (column Original) and understood by our annotators in the last round of the coping annotation phase of crowdsourcing. All annotators in this round identified the correct coping strategy (Correct Annotators). ✓ means that an emotion average rating was > 2. Numbers are rankings of emotion averages.

Emotions	Distance			Reject		
	Original	Correct Annotat.	Wrong Annotat.	Original	Correct Annotat.	Wrong Annotat.
Anger					✓, 2	
Contempt				✓	✓, 4	✓, 3
Disgust				✓	✓, 3	✓, 4
Dislike	✓	✓, 3	✓, 1		✓, 1	✓, 1
Distress	✓	✓, 1	✓, 2		✓, 5	
Fear	✓	✓, 2	✓, 3			
Frustration		✓, 5	✓, 4		✓, 6	✓, 2
Guilt						
Hope						
Joy						
Love						
Pride						
Regret	✓		✓, 6			
Relief						
Sadness	✓	✓, 4	✓, 5			
Shame			✓, 7	✓		
Surprise						

Table 10: Emotions for the strategy Distance and Reject as indicated by [Roseman \(2013\)](#) (column Original) and understood by our annotators in the last round of the coping annotation phase of crowdsourcing. Annotators are divided into those who identified the correct coping strategy (column Correct Annot.) and those who failed the task (column Wrong Annot.). ✓ means that an emotion average rating was > 2. Numbers are rankings of emotion averages.

B Generating Scenarios

We identified the topics via ChatGPT-4, asked to list 20 sensitive topics for specific categories of people. This step was independently performed by three researchers, who later integrated and refined their outputs in a set of 5 topics: abortion, immigration, racism, LGBTQ+, drugs.

For the scenario generation, ChatGPT-4 had to refrain from characterizing the gender of the interlocutors (hence, the names X and Y). Scenarios were produced between August 14 and October 31 (2023) with ChatGPT-4.

Table 11 shows how we prompted ChatGPT-4 (we show it for only one topic and one scenario for brevity) and Table 12 reports all scenarios we used in our study.

Table 11: ChatGPT-4 generation of scenarios. Green rows correspond to humans, white rows to the assistant's replies. For simplicity, we only report one generated scenario.

I am conducting a study about Roseman's coping strategies (i.e., attack, contact, distance and reject). My goal is to study how each strategy realises in language when speakers discuss about controversial topics. Imagine that during an interaction, one of the interlocutors expresses an opinion that makes another uncomfortable; the latter thus responds with a verbal utterance and a non-verbal reaction while being "under the influence" of a coping strategy.

Certainly! Studying how Roseman's coping strategies manifest in language, especially during discussions on controversial topics, is an intriguing area of inquiry. When individuals discuss controversial topics and face conflicting or uncomfortable opinions, they might adopt one or more of Roseman's coping strategies in their verbal and non-verbal behavior.

Here's a breakdown of each strategy along with potential verbal and non-verbal manifestations:

1. **Attack:**

Verbal: An individual might use aggressive language, accusations, or challenges. They might say things like, "That's completely wrong," "How can you even believe that?", or "You clearly haven't done your research."

Non-Verbal: Physically leaning forward, intense eye contact, clenched fists, raised voice, or even pointing fingers. The individual's face might show signs of anger or frustration.

2. **Contact:**

Verbal: The individual would seek clarification, ask questions, or try to understand the other's viewpoint. They might say, "Can you explain why you feel that way?", "I'm curious to know more about your perspective," or "Let's find some common ground."

Non-Verbal: Leaning in, maintaining steady but non-threatening eye contact, nodding to indicate active listening, and open body posture. The face might reflect genuine curiosity or concern.

3. **DISTANCE:**

Verbal: An individual might divert the topic, use vague language, or minimize their involvement in the conversation. They might say, "Let's discuss this later," "I need some time to think about it," or simply change the topic altogether.

Non-Verbal: Looking away, physically leaning back, creating physical distance, crossing arms as a defensive posture, or checking one's phone/watch. The face might show signs of discomfort or avoidance.

4. **Reject:**

Verbal: The individual might dismiss the topic or the person's opinion outright, often without offering a counter-argument. They might say, "That's not even worth discussing," "I can't entertain that thought," or "We'll never agree on this, so let's drop it."

Non-Verbal: Eye-rolling, shaking the head, waving one's hand dismissively, or even walking away. The face might display contempt, disdain, or indifference.

For a comprehensive study, you may also want to consider the tone of voice, the specific linguistic constructions used, the rhythm and pace of speech, and other nuanced non-verbal cues like pupil dilation or micro-expressions. Moreover, always remember that cultural and individual variations can affect how these coping strategies manifest. It would be beneficial to take these variations into account when analyzing and interpreting your findings.

With my next prompts, I'm going to ask you to conduct the following task.

Task Depict three scenarios (each set in a different social context, briefly described) in which two characters, X and Y, interact. Y produces a controversial statement about a given topic that I will specify. That opinion could elicit an emotional reaction in X. Each scenario must be structured like so:

- SCENARIO AND CONTROVERSIAL OPINION
- VERBALIZED REACTION AND NON VERBALIZED REACTION ASSUMING THAT PERSON B ADOPTS THE COPING STRATEGY OF ATTACK
- VERBALIZED REACTION AND NON VERBALIZED REACTION ASSUMING THAT PERSON B ADOPTS THE COPING STRATEGY OF CONTACT
- VERBALIZED REACTION AND NON VERBALIZED REACTION ASSUMING THAT PERSON B ADOPTS THE COPING STRATEGY OF DISTANCE - VERBALIZED REACTION AND NON VERBALIZED REACTION ASSUMING THAT PERSON B ADOPTS THE COPING STRATEGY OF REJECT

Requirements:

- SCENARIO AND CONTROVERSIAL OPINIONS should span maximum 6 sentences;
- for each scenario, the controversial statement must focus on a different aspect of the issue;
- X must have a verbalized and non-verbalized reaction corresponding to one coping strategy;
- each scenario should feature a different power relationship between X and Y, with them being peers in the first scenario (none holds more power than the other), X being the dominant individual in the second, and Y being the dominant individual in the third;
- when describing the scenario, do not mention the demographics of the interlocutors, but specify their role to one another (e.g., who is Y to X?) as to make the reader understand their relationship.

I will later input the topics and you will generate the scenarios. For now, just confirm if the task is clear.

The task is clear. Once you provide the topics for the controversial statements, I will generate scenarios based on the structure and requirements you've outlined. Just to recap:

1. We'll have three scenarios set in distinct social contexts.
 2. The nature of the relationship between X and Y will differ in each scenario: peers in the first, X dominant in the second, and Y dominant in the third.
 3. Y will make a controversial statement in each scenario.
 4. X's reaction will be provided for each of the four coping strategies, both verbally and non-verbally.
 5. The demographic details of the interlocutors won't be specified, but their roles/relationship will be.
- Please provide the topics for the controversial statements, and I'll proceed with creating the scenarios.

Please generate three different scenarios for the topic immigration policy and border control (at least one of them should focus on islamophobia).

SCENARIO 1: Colleagues at a Company Coffee Break

SCENARIO AND CONTROVERSIAL OPINION X and Y are colleagues who work in the same department and have known each other for a year. During a coffee break, they discuss current events. Y, stirring their coffee, remarks, "Honestly, I think our country's immigration policy is too lenient. We're just letting anyone in, and it's bound to create chaos."

- **ATTACK**

Verbalised Reaction "That's such a narrow-minded view, Y! It's people like you who perpetuate unnecessary fear."

Non-Verbalised Reaction X's eyes narrow, lips tighten, leans forward aggressively, fists clenching the edge of the table.

- **CONTACT**

Verbalised Reaction "Why do you feel that way, Y? Do you think a stricter policy would really solve the perceived problems?"

Non-Verbalised Reaction X tilts head slightly, maintains steady eye contact, eyebrows raised in a questioning manner.

- **DISTANCE**

Verbalised Reaction "Hmm, interesting perspective. Have you tried the new blend of coffee they have here?"

Non-Verbalised Reaction X takes a long sip from their mug, looks away briefly, and shifts weight from one foot to the other.

- **REJECT**

Verbalised Reaction "I don't think this is the place or time to discuss such matters. Let's get back to work."

Non-Verbalised Reaction X sighs, rolls eyes discreetly, and starts gathering their things.

Table 12: Scenarios used in the role-playing phase of data creation, divided by topic.

Immigration	
X-Y Relationship	X and Y are colleagues who work in the same department and have known each other for a year.
Scenario 1	During a coffee break, they discuss current events. Y, stirring its coffee, remarks, “Honestly, I think our country’s immigration policy is too lenient. We’re just letting anyone in, and it’s bound to create chaos.”
X-Y Relationship	Y is a student who stays back after class to discuss an assignment with X, the teacher.
Scenario 2	During their conversation, Y says, “It’s obvious that the recent surge in Muslim immigrants is just a strategy for them to dominate our culture.”
X-Y Relationship	X is an applicant for a managerial role, while Y is the CEO of the company conducting the interview.
Scenario 3	As they discuss potential challenges, Y mentions, “We’ve had issues with foreign workers not blending in. Honestly, open borders are more of a problem than a solution.”
Racism	
X-Y Relationship	X is a classmate to Y
Scenario 1	During a university class discussion on historical racial events, B confidently states, “People keep talking about systemic racism, but I believe that’s just an excuse for those who don’t want to work hard. If you look around, everyone has the same opportunities today.”
X-Y Relationship	X is an instructor, and Y is a regular attendee of a fitness class.
Scenario 2	During a water break, Y comments to a friend, but loud enough for X to hear, “I still can’t believe they hired a Latino as our instructor.”
X-Y Relationship	X and Y are siblings, having dinner at their parents’ home.
Scenario 3	As they discuss friends and relationships, Y says, “You know I’ve always believed people should stick to their own kind. It’s just easier.”
Drugs	
X-Y Relationship	X is a concerned parent, and Y is a local politician.
Scenario 1	During a town hall meeting, Y proposes, “Legalizing all drugs would cut down on crime and help manage addiction better through official channels.”
X-Y Relationship	X is a podcast host, and Y is an expert in public health.
Scenario 2	During the recording of a podcast discussion, Y mentions, “The ‘War on Drugs’ has largely been ineffective. A harm reduction approach is more viable.”
X-Y Relationship	X and Y are friends.
Scenario 3	X and Y are friends catching up over coffee. As the topic shifts, Y shares, “I’ve been reading about how some countries are decriminalizing drugs, and it seems like a good move. It focuses on treating users instead of punishing them.”
Abortion	
X-Y Relationship	X and Y are both volunteers at a local food drive.
Scenario 1	While sorting through donations, Y mentions, “Honestly, limiting access to abortion is just another way of controlling women. Everyone should have a say about their own bodies.”
X-Y Relationship	X is an attendee at a charity gala, and Y is the guest speaker.

Scenario 2:	In Y's speech about women's health, they assert, "It's time to prioritize the rights of women by ensuring unrestricted access to abortion services."
X-Y Relationship Scenario 3	At a charity gala dinner, X is a guest, and Y is the keynote speaker. During the speech, Y mentions, "In a world where we're fighting for so many rights, I wish more emphasis was placed on the rights of the unborn."
LGBTQ+	
X-Y Relationship Scenario 1	X and Y are siblings, sitting together at a large family gathering. While discussing recent events, Y comments, "I don't get this whole LGBTQ+ thing. There are just two genders. Why complicate things?"
X-Y Relationship Scenario 2	X is a department manager, and Y is a colleague from a different department. While having coffee, Y says, "These workplace LGBTQ+ training sessions are so unnecessary. Why can't people just keep their personal lives out of work?"
X-Y Relationship Scenario 3	X is the class president, and Y is an old classmate. During the reunion, Y whispers to X, "Have you noticed? There are so many gay couples here now. It wasn't like this during our school days."

C Role-Playing

While we assume that the participants can simulate coping strategies and propagate them in text, doing that with laypeople in an artificial setup is challenging. We therefore conducted the task in different rounds, to experiment with different implementations of the role-playing task. In the first round, we assigned participants to one X to impersonate. Observing that this setup complicated the role-playing performance, in the following rounds we assigned participants to the X that they mapped to the right s in t : we showed them the 4 X 's descriptions and allowed them to proceed with the subsequent questions based on the X that they label correctly.

Concerning the participants, we opened the study to crowdworkers with a Prolific approval rate between 90% and 100%. We detail the requirements for participation in Table 13, divided by topic (the cross symbol in the table means we did not use a given filter). At each round, we excluded participants from previous rounds. For most studies, we selected participants with English as their first language, and coming from the following countries: UK, US, Canada, Ireland, New Zealand, Australia. The asterisk means we use only a subset.

We reject participants who did not write in first person, did not use direct speech, who produced ungrammatical or too short text, did not disagree with Y , or used too little time to answer (less than 3 minutes).

Topic	Allowed Participants
Immigration [†]	Immigrated to the country where currently living. Place where spent most time before turning 18: Poland, Germany, India, Pakistan, Ireland, Romania, Mexico, China, Philippines, Dominican Republic, Cuba (i.e., the countries where majority of immigrants in UK and US come from).
Racism	Black, Asian, Mixed, Other
Drugs	Have taken part in alcohol therapy
Abortion	Women, Pro choice and Pro life
LGBTQ+	$65 \leq$ homosexual, bisexual, asexual, other (non heterosexual)
Healthcare	Have a chronic disease
Ageism	$65 \leq \text{Age} \leq 100$

Table 13: Filters used in Prolific to allow participants in our studies. * means we use only a subset of the basic set of residence countries (UK, US, Ireland, New Zealand, Australia), and [†] means we did not use English as first language.

Attack	think, drugs, rights, people, don, women, right, abortion, disagree, say
Contact	people, think, drugs, believe, different, say, agree, way, good, don
Distance	think, people, drugs, don, say, want, disagree, agree, rights, life
Reject	don, think, people, drugs, rights, want, say, need, believe, unborn

Table 14: Most common words per coping strategy, using tf-idf (to be redone, this was not found on all data).

D Corpus Analysis

Table 14 reports the 10 words with the highest tf-idf across coping strategies. Figure 5 reports an example comparison between the emotions assigned to character *X* and the same emotion for respondents who claimed their own reaction to *Y* would be different from *X*’s.

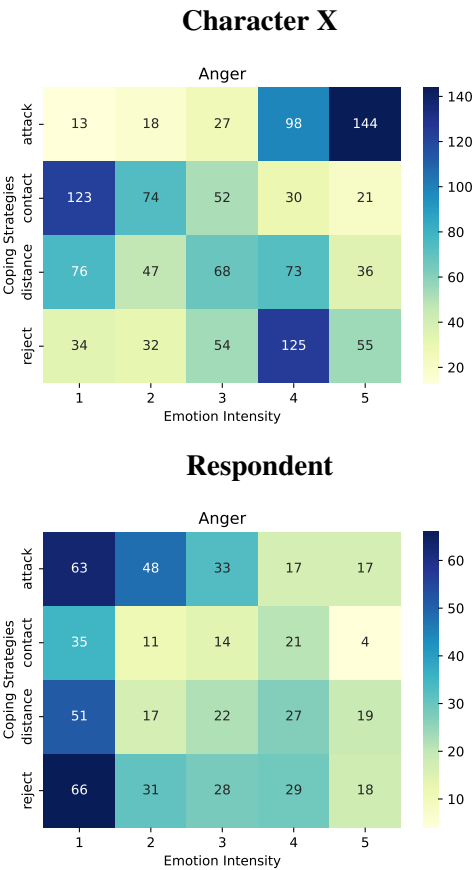


Figure 5: Comparison of anger scores assigned to character *X* and as self-assigned by participants who would not react as *X*. Numbers in each cell are the numbers of participants picking a given emotion score (intensity).

E Classification Experiments

We conduct zero-shot classification experiments on our local machines, with the quantized models PHI-2.Q8_0.GGUF, MISTRAL-7B-INSTRUCT-v0.2.Q8_0.GGUF, DOLPHIN-2.0-MISTRAL-7B.Q5_K_M.GGUF, and NOUS-HERMES-2-SOLAR-10.7B.Q5_K_M.GGUF, all downloaded from <https://huggingface.co/TheBloke>.

Table 15 reports a structured difference between the prompts (P1, P2, and P3) used in the zero-shot classification experiments. Each prompt was used in three setups, i.e., by concatenating the task instructions with X 's answer only, with the description of X 's behavior only, or with a concatenation of X 's answer and behavior.

Results obtained with all prompts are in Table 16. F_1 scores reported in Table 6 correspond to P3.

	Prompt 1	Prompt 2	Prompt 3
Introduction	People respond to emotional situations with four different strategies.	People can oppose a view by adopting one of four emotional attitudes.	According to Roseman, there exist four emotion coping strategies, i.e., attack, contact, distance and reject.
Definitions	Attack: the experiencer responds in an aggressive manner. Contact: the experiencer responds in an understanding manner. Distance: the experiencer seeks personal space away from the stimulus. Reject: the experiencer promptly refuses the stimulus.	Attack: to right the wrong in an aggressive manner. Contact: to right the wrong with an open mindset. Distance: to seek shelter from unpleasant situations in a quiet inner space. Reject: to approach life in a firm refusal of negativity.	Attack: coming across as aggressive and combative, facing conflicts fiercely, taking action to right the wrong. Contact: acknowledging the diversity of opinions, responding with a constructive attitude, trying to solve problems in a respectful manner. Distance: seeking emotional tranquillity and self-reflection, coming across as avoidant, disengaging from conflicts. Reject: judgmental attitude, resolute attempt to refuse unpleasant situations, expressing opinions with confidence, dismissing potential conflicts.
Task	Please classify the following text with the one strategy being implemented in ...	Please classify the following text with the one attitude emerging from	Given these strategies, please recognize what strategy is being used by the ...
Answer	...it (choose one and only one label).	...it (choose one and only one label).	...utterer of the following text (choose one and only one label). ...
Behavior	...the described behavior (choose one and only one label) – note: the text is about how person X behaves in an unpleasant situation, using a coping strategy.	...the described behavior (choose one and only one label) – note: the text is about how person X behaves in an unpleasant situation, using a coping strategy.	...person behaving as described in the following text (choose one and only one label) – note: the text is about how person X behaves in an unpleasant situation, using a coping strategy.
Answer and Behavior	...it (choose one and only one label) – note: what follows includes a verbal reaction to an unpleasant situation and a description of how the utterer behaves in that context.	...it (choose one and only one label) – note: what follows includes a verbal reaction to an unpleasant situation and a description of how the utterer behaves in that context.	...utterer of the following text (choose one and only one label) – note: what follows includes a verbal reaction to an unpleasant situation and a description of how the utterer behaves in that context.

Table 15: Zero-shot setup prompts, obtained by concatenating the Introduction, the coping strategies Definitions, and one labeling Task either focusing on X ’s answer, X ’s behavior or a combination of the two.

	Answer			Behavior			Answ.+Behav.		
	P1	P2	P3	P1	P2	P3	P1	P2	P3
phi-2	.345	.307	.356	.338	.350	.391	.397	.321	.322
mistral-2	.380	.379	.392	.473	.520	.474	.495	.476	.538
dolphin-2	.385	.376	.422	.449	.513	.550	.454	.448	.520
nous-hermes	.429	.480	.387	.482	.518	.526	.523	.529	.515

Table 16: F1 with different prompts, with X ’s answer as input (Answer), with the description of X ’s behavior (Behavior), or a concatenation of both (Answ.+Behav.). Each of these input types were separately concatenated with three prompts (P1, P2, and P3), fed to the four models (rows).